

Final Project Report

Incident Life Cycle Automation - consolidated project report

DATE	STUDENT NAME	PROJECT
07 October 2026	Gangadevi.R	Incident Life Cycle Automation

1. Introduction

1.1 Project Overview

Incident Life Cycle Automation is a workflow-oriented system designed to manage operational incidents as one continuous, traceable lifecycle. The project focuses on reducing manual coordination by automating intake, triage, assignment, SLA monitoring, escalation, resolution and closure.

1.2 Purpose

The purpose is to improve response visibility, ownership, consistency and auditability without introducing unnecessary scope beyond the incident lifecycle itself.

2. Ideation Phase

2.1 Problem Statement

Incident handling can be fragmented across manual steps, leading to delayed ownership, unclear status, missed SLA actions and incomplete history.

2.2 Empathy Map Summary

- Users need a simple reporting path and transparent status.
- Support teams need complete incident context and clear ownership.
- Managers need early SLA risk visibility and escalation control.
- Auditors need reliable lifecycle history.

2.3 Brainstorming

The selected initial scope prioritizes the highest-value, feasible ideas: structured intake, automated routing, SLA monitoring, escalation, lifecycle status, resolution and audit history.

3. Requirement Analysis

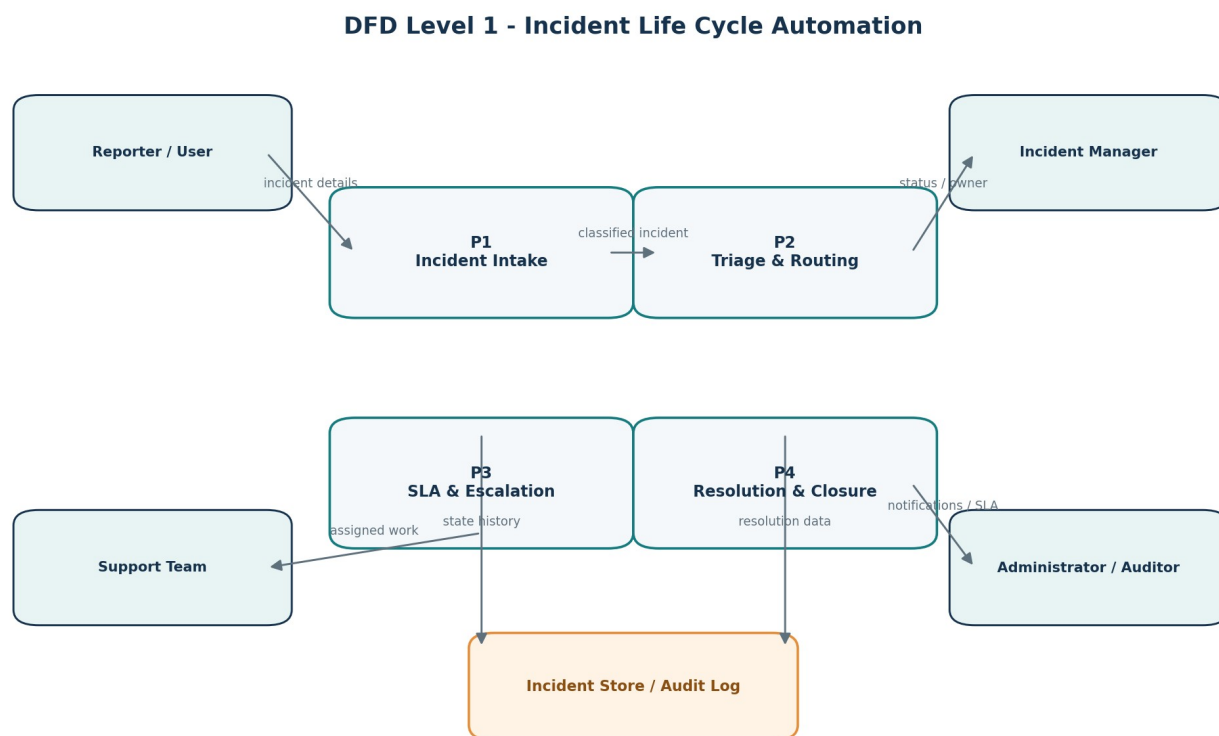
3.1 Customer Journey Map

The target journey is: notice issue -> report -> receive ownership -> track progress -> respond to escalation -> confirm resolution -> close and review.

3.2 Solution Requirements

Core requirements cover incident intake, identification, classification, assignment, SLA tracking, notifications, investigation, resolution, closure, history, dashboard/search, access control and auditability.

3.3 Data Flow Diagram



3.4 Technology Stack

Proposed stack: HTML/CSS/JavaScript for the interface, Python/Flask for application logic, SQLite/MySQL for persistence, a background scheduler/worker for SLA checks, and SMTP/email for notifications. This is a proposed project stack and should be aligned with the actual implementation before submission.

4. Project Design

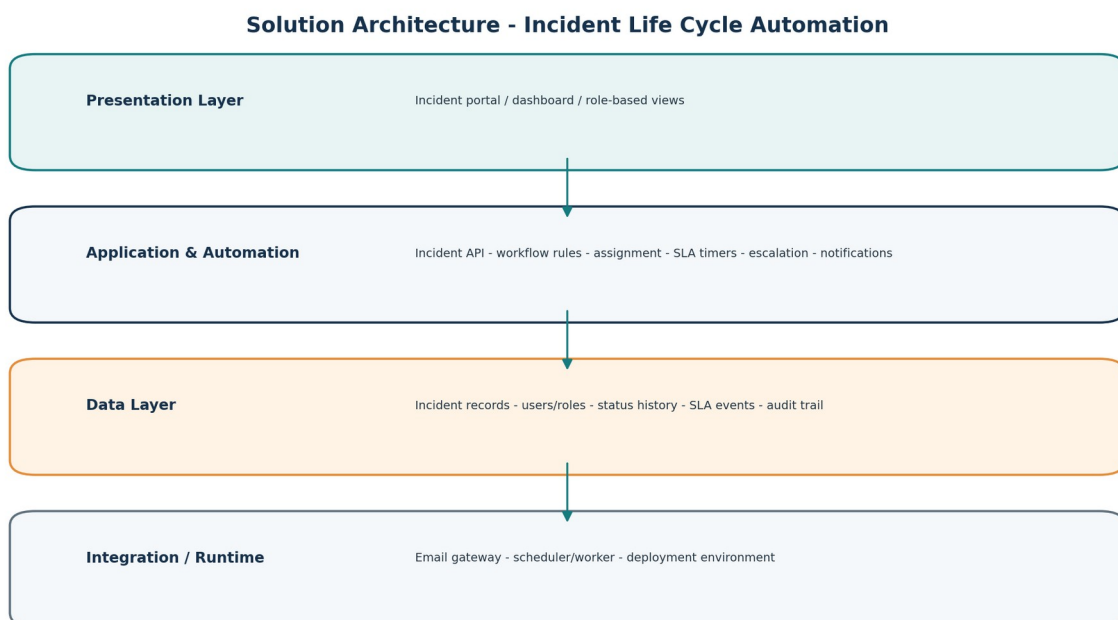
4.1 Problem-Solution Fit

Each core automation directly addresses an observed lifecycle pain: structured intake reduces missing details, routing reduces ownership delay, SLA automation reduces late follow-up, and audit history reduces traceability gaps.

4.2 Proposed Solution

The solution automates incident intake, rule-based triage and assignment, SLA tracking, escalation, resolution, closure and lifecycle history. Each transition is time-stamped so ownership and status remain visible throughout the incident.

4.3 Solution Architecture



5. Project Planning & Scheduling

5.1 Project Plan

Sprint	Focus	Planned Points
Sprint 1	Incident foundation - intake, triage, priority, assignment	16
Sprint 2	Active handling - status, investigation, resolution, SLA	13
Sprint 3	Operational visibility - dashboard, rules, audit, notifications	13

6. Functional & Performance Testing

6.1 Functional Testing

Functional coverage includes positive and negative incident creation, assignment, status changes, SLA behavior, resolution, closure, search and role-based actions.

6.2 Performance Testing

Performance checks measure incident creation, detail retrieval, search/filtering, scheduled SLA processing and concurrent updates. Actual timings must be captured from the implemented system rather than assumed.

6.3 User Acceptance Testing

UAT validates that the complete user workflow matches expected operational behavior and records any defects or sign-off decisions.

7. Results

The final results section should contain screenshots from the implemented application, including: incident creation, incident detail/timeline, assignment, SLA state, dashboard/search and closure/audit history. Attach measured test outcomes from execution.

Implementation Screenshot 1

ServiceNow Developers INC0010004 | Service Operat Corporate VPN | Offering | Remote Access | Service | Ser (19) WhatsApp Ask Gemini

dev449470.service-now.com/cmdb_ci_service.do?sys_id=542a63c3836f031092de5dc6feaad3c3&sysparm_record_target=cmdb_ci_service&sysparm_record_row=1&sysparm_record_rows=22&sys...

Service Remote Access

Open in CMDB Workspace Update Convert to Service Instance Delete

Name Remote Access

Model ID

Owned by

Business criticality 2 - somewhat critical

Version

Used for Production

Operational status Operational

Service classification Business Service

Comments

Approval group

Support group

Change Group

Managed by

SLA

Location

Vendor

Related Items

Search for CI

Open in CMDB Workspace Update Convert to Service Instance Delete

Related Links

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Implementation Screenshot 2

ServiceNow Developers INC0010004 | Service Operat Corporate VPN | Offering | Remote Access | Service | Ser (19) WhatsApp Ask Gemini

dev449470.service-now.com/service_offering.do?sys_id=767a6fc3836f031092de5dc6feaad371&sysparm_record_target=service_offering&sysparm_record_row=1&sysparm_record_rows=1&sys...

Offering Corporate VPN

Open in CMDB Workspace Update Delete

Name Corporate VPN

Model ID

Owned by

* Parent Remote Access

Used for Production

Operational status Operational

Vendor

Approval group

Support group

Change group

Managed by

Version

Business criticality 2 - somewhat critical

SLA

Location

Short Description

Description

Comments

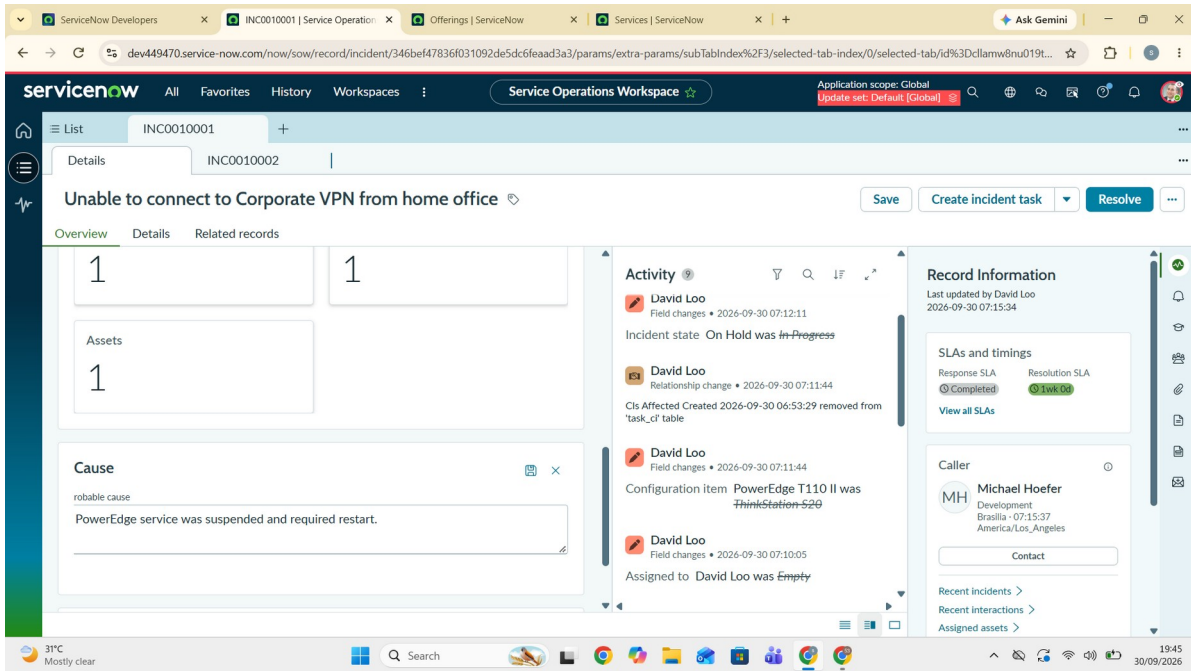
Related Items

Show 3 Levels

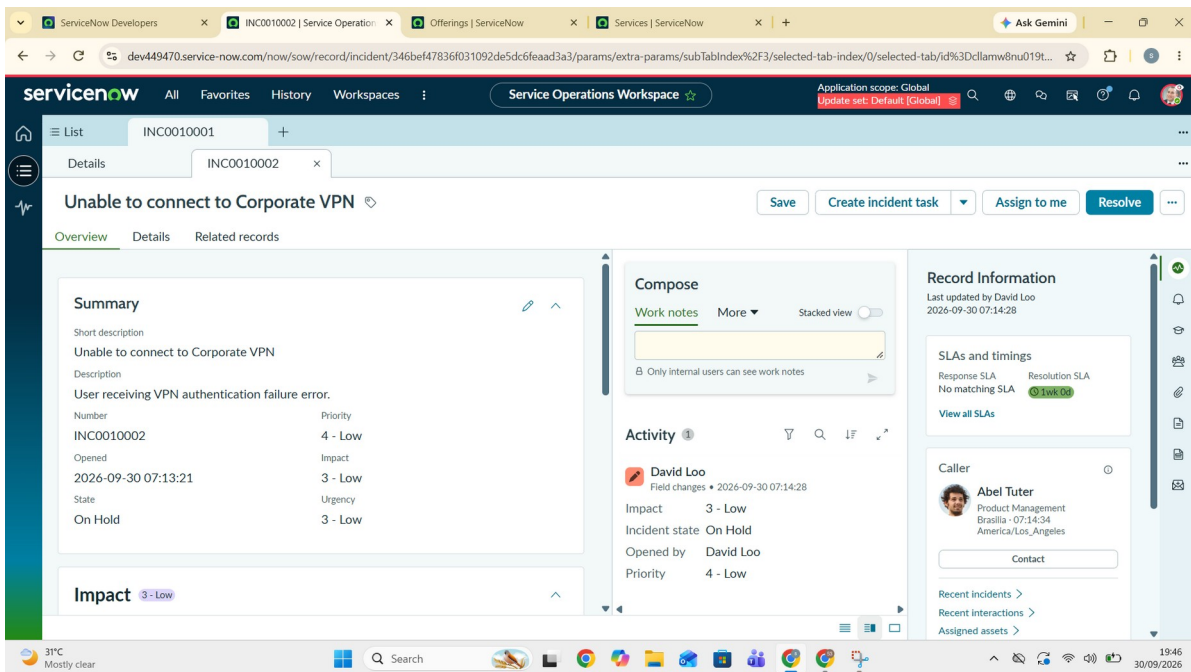
Contract

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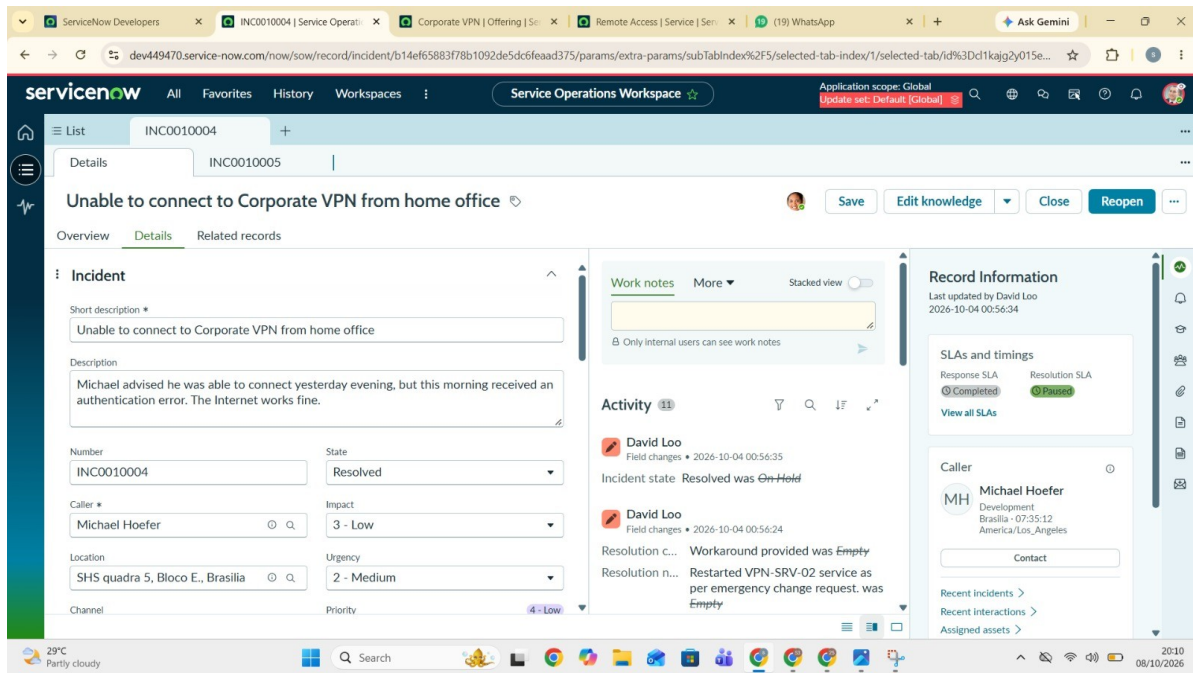
Implementation Screenshot 3



Implementation Screenshot 4



Implementation Screenshot 5



8. Advantages & Limitations

Advantages	Limitations
Clear ownership and lifecycle visibility	Initial scope is limited to core incident automation
Reduced repetitive manual follow-up	Advanced integrations are outside the initial build
SLA-driven escalation	Performance depends on deployment environment
Consistent status history and audit trail	Rules require correct configuration

9. Conclusion

Incident Life Cycle Automation converts a fragmented incident-handling process into a rule-driven workflow. The project is designed to improve operational visibility, reduce manual coordination and preserve a reliable history from incident creation through closure.

10. Future Scope

- Service-specific SLA policies and more granular workflow rules
- External ticketing / email / messaging integrations
- Analytics for incident trends and recurring failure patterns

11. Appendix

- Source code: attach project repository or source archive.
- Project screenshots: attach implementation screenshots.

- Test evidence: attach executed UAT and performance results.
- Demo link: add only if an actual deployment exists.